



In this chapter you will learn about...

- ☐ Fractions
- ☐ Comparison of fractions
- ☐ Lowest terms of a fraction
- ☐ Addition and subtraction of fractions
- ☐ Multiplication of fractions
- ☐ Multiplicative inverse or reciprocal of a fraction
- ☐ Division of fractions

FRACTIONS

The number of the form $\frac{a}{b}$, where $b \neq 0$ and a, b are whole numbers is called a **fraction**. Here, a is called the **numerator** and b is called the **denominator**.

Types of Fractions

- (i) **Proper fraction** : If the numerator is smaller than the denominator, then the fraction is called **proper fraction**. *Examples* : $\frac{1}{3}, \frac{4}{5}, \frac{2}{7}$ etc.
- (ii) **Improper fraction** : If the numerator is greater than or equal to the denominator, then the fraction is called **improper fraction**. *Examples* : $\frac{5}{3}, \frac{7}{2}, \frac{6}{5}$ etc.
- (iii) **Like fractions** : Fractions with same denominators are called **like fractions**.
Examples : $\frac{2}{7}, \frac{3}{7}, \frac{5}{7}, \frac{6}{7}$ etc.
- (iv) **Unlike fractions** : Fractions with different denominators are called **unlike fractions**.
Examples : $\frac{1}{2}, \frac{5}{7}, \frac{4}{9}$ etc.
- (v) **Unit fractions** : Fractions whose numerator is always '1' are called **unit fractions**.
Examples : $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}$ etc.
- (vi) **Mixed fractions** : Fractions having whole number along with proper fraction are known as **mixed fractions**. *Examples* : $1\frac{2}{3}, 1\frac{4}{5}, 2\frac{2}{3}$ etc.

(vii) **Decimal fractions** : Fractions having denominators as 10, 100, 1000, etc. are called decimal fractions. *Examples* : $\frac{7}{10}, \frac{3}{100}, \frac{41}{1000}$ etc.

(viii) **Equivalent fractions** : A given fraction and fraction obtained by multiplying or dividing its numerator and denominator by the same non-zero number are called equivalent fractions.

Examples : $\frac{2}{3}, \frac{4}{6}, \frac{6}{9}, \frac{8}{12}$ are equivalent fractions.

(ix) **Vulgar fractions** : A fraction whose denominator is a whole number, other than 10, 100, 1000 etc. is called a vulgar fraction. *Examples* : $\frac{7}{8}, \frac{3}{17}, \frac{19}{20}, \frac{23}{39}$ etc.

Example 1 : Convert each of the following into an improper fraction :

(i) $1\frac{4}{5}$

(ii) $3\frac{2}{7}$

(iii)

(iv) $12\frac{3}{5}$

Solution : (i) $1\frac{4}{5} = \frac{(1 \times 5) + 4}{5} = \frac{5 + 4}{5} = \frac{9}{5}$

(ii) $3\frac{2}{7} = \frac{(3 \times 7) + 2}{7} = \frac{21 + 2}{7} = \frac{23}{7}$

(iii) $4\frac{2}{3} = \frac{(4 \times 3) + 2}{3} = \frac{12 + 2}{3} = \frac{14}{3}$

(iv) $12\frac{3}{5} = \frac{(12 \times 5) + 3}{5} = \frac{60 + 3}{5} = \frac{63}{5}$

Example 2 : Convert each of the following into a mixed fraction :

(i) $\frac{49}{5}$

(ii) $\frac{68}{9}$

(iii) $\frac{95}{14}$

Solution : (i) On dividing 49 by 5, we get quotient as 9 and remainder as 4.

$\therefore \frac{49}{5} = 9\frac{4}{5}$

$$\begin{array}{r} 9 \\ 5 \overline{) 49} \\ \underline{-45} \\ 4 \end{array}$$

(ii) On dividing 68 by 9, we get quotient as 7 and remainder as 5.

$\therefore \frac{68}{9} = 7\frac{5}{9}$

$$\begin{array}{r} 7 \\ 9 \overline{) 68} \\ \underline{-63} \\ 5 \end{array}$$

(iii) On dividing 95 by 14, we get quotient as 6 and remainder as 11.

$\therefore \frac{95}{14} = 6\frac{11}{14}$

$$\begin{array}{r} 6 \\ 14 \overline{) 95} \\ \underline{-84} \\ 11 \end{array}$$

COMPARISON OF FRACTIONS

Rule 1 : When the fractions have same denominators, the fraction with larger numerator is larger.

Examples : $\frac{9}{11} > \frac{7}{11}, \frac{3}{8} > \frac{1}{8}, \frac{5}{7} > \frac{3}{7}$

Rule 2 : When the fractions have same numerators, the fraction with larger denominator is smaller.

Examples : $\frac{3}{7} < \frac{3}{5}$, $\frac{7}{11} < \frac{7}{9}$, $\frac{5}{14} < \frac{5}{12}$

Rule 3 : When the fractions have different denominators and numerators, then we multiply or divide the fractions with same number to make denominator same and then apply Rule 1.

Example : Which is greater : $\frac{3}{5}$ or $\frac{7}{8}$?

Solution : L.C.M. of denominators 5 and 8 is 40.

$$\therefore \frac{3}{5} = \frac{3 \times 8}{5 \times 8} = \frac{24}{40}; \frac{7}{8} = \frac{7 \times 5}{8 \times 5} = \frac{35}{40}$$

$$\text{Now, } \frac{35}{40} > \frac{24}{40}$$

$$\text{Hence, } \frac{7}{8} > \frac{3}{5}$$

Alternative Method : (Cross Multiplication Method)

To compare $\frac{3}{5}$ and $\frac{7}{8}$ by cross multiplication method, we write

$$\begin{array}{ccc} \frac{3}{5} & \times & \frac{7}{8} \\ 3 \times 8 & & 7 \times 5 \\ 24 & & 35 \end{array}$$

$$\text{Since, } 35 > 24$$

$$\text{Hence, } \frac{7}{8} > \frac{3}{5}$$

Note : In general if $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then

$$(i) \quad \frac{a}{b} > \frac{c}{d} \Leftrightarrow ad > bc.$$

$$(ii) \quad \frac{a}{b} = \frac{c}{d} \Leftrightarrow ad = bc.$$

$$(iii) \quad \frac{a}{b} < \frac{c}{d} \Leftrightarrow ad < bc.$$

LOWEST TERMS OF A FRACTION

A fraction $\frac{a}{b}$ is said to be in lowest terms (irreducible) if H.C.F. of a and b is 1.

Any fraction may be reduced to lowest terms by dividing the numerator and denominator by their H.C.F.

Example 1 : Reduce the following fractions to their lowest terms :

$$(i) \quad \frac{78}{338}$$

$$(ii) \quad \frac{84}{144}$$

$$(iii) \quad \frac{105}{270}$$

Solution : (i) H.C.F. of 78 and 338 is 26.

$$\therefore \frac{78}{338} = \frac{78 \div 26}{338 \div 26} = \frac{3}{13}$$

$\frac{78}{338}$ in its lowest terms is equal to $\frac{3}{13}$.

$$\begin{array}{r} 4 \\ 78 \overline{) 338} \\ \underline{-312} \\ 26 \overline{) 78} (3 \\ \underline{-78} \\ 0 \end{array}$$

(ii) H.C.F. of 84 and 144 is 12.

$$\therefore \frac{84}{144} = \frac{84 \div 12}{144 \div 12} = \frac{7}{12}$$

$\frac{84}{144}$ in its lowest terms is equal to $\frac{7}{12}$.

$$\begin{array}{r} 1 \\ 84 \overline{) 144} \\ \underline{-84} \\ 60 \overline{) 84} (1 \\ \underline{-60} \\ 24 \overline{) 60} (2 \\ \underline{-48} \\ 12 \overline{) 24} (2 \\ \underline{-24} \\ 0 \end{array}$$

(iii) H.C.F. of 105 and 270 is 15.

$$\therefore \frac{105}{270} = \frac{105 \div 15}{270 \div 15} = \frac{7}{18}$$

$\frac{105}{270}$ in its lowest terms is equal to $\frac{7}{18}$.

$$\begin{array}{r} 2 \\ 105 \overline{) 270} \\ \underline{-210} \\ 60 \overline{) 105} (1 \\ \underline{-60} \\ 45 \overline{) 60} (1 \\ \underline{-45} \\ 15 \overline{) 45} (3 \\ \underline{-45} \\ 0 \end{array}$$

Example 2 : Arrange the following fractions in ascending order :

$$\frac{1}{3}, \frac{2}{5}, \frac{5}{6}, \frac{1}{15}$$

Solution : L.C.M. of 3, 5, 6 and 15 = $3 \times 5 \times 2 = 30$

$$\begin{array}{c|c} 3 & 3, 5, 6, 15 \\ 5 & 1, 5, 2, 5 \\ \hline & 1, 1, 2, 1 \end{array}$$

$$\frac{1}{3} = \frac{1 \times 10}{3 \times 10} = \frac{10}{30}, \quad \frac{2}{5} = \frac{2 \times 6}{5 \times 6} = \frac{12}{30}$$

$$\frac{5}{6} = \frac{5 \times 5}{6 \times 5} = \frac{25}{30}, \quad \frac{1}{15} = \frac{1 \times 2}{15 \times 2} = \frac{2}{30}$$

$\therefore$ The denominators are same, so the fractions having the greater numerator is greater.

$\therefore \frac{2}{30}, \frac{10}{30}, \frac{12}{30}, \frac{25}{30}$ are in ascending order.

$\Rightarrow \frac{1}{15}, \frac{1}{3}, \frac{2}{5}, \frac{5}{6}$ are in ascending order.



EXERCISE 2.1

1. Write each of the following fractions into lowest terms :

(i) $\frac{18}{48}$

(ii) $\frac{120}{168}$

(iii) $\frac{45}{180}$

(iv) $\frac{81}{108}$

2. Convert each of the following into mixed fractions :

(i) $\frac{16}{3}$

(ii) $\frac{32}{7}$

(iii) $\frac{243}{13}$

(iv) $\frac{441}{17}$

3. Arrange in descending order :

(i) $\frac{5}{27}, \frac{4}{9}, \frac{7}{24}, \frac{5}{12}$

(ii) $\frac{3}{8}, \frac{5}{14}, \frac{3}{4}, \frac{5}{7}$

4. Arrange in ascending order :

(i) $\frac{3}{4}, \frac{7}{12}, \frac{5}{6}, \frac{11}{12}$

(ii) $\frac{4}{5}, \frac{8}{15}, \frac{7}{10}, \frac{13}{20}$

(iii) $\frac{4}{5}, \frac{9}{14}, \frac{16}{35}, \frac{6}{10}$

(iv) $\frac{1}{5}, \frac{7}{10}, \frac{3}{7}, \frac{2}{15}$

5. Convert each of the following into improper fraction :

(i) $2\frac{3}{4}$

(ii) $3\frac{3}{10}$

(iii) $6\frac{2}{3}$

(iv) $2\frac{5}{6}$

(v) $6\frac{2}{5}$

6. Which of the following fractions are equivalent?

(i) $\frac{9}{15}$ and $\frac{3}{5}$

(ii) $\frac{7}{21}$ and $\frac{3}{9}$

(iii) $\frac{5}{7}$ and $\frac{15}{28}$

(iv) $\frac{6}{13}$ and $\frac{30}{52}$

(v) $\frac{4}{9}$ and $\frac{20}{45}$

(vi) $\frac{8}{36}$ and $\frac{2}{9}$

ADDITION AND SUBTRACTION OF FRACTIONS

Rule 1 : In order to add or subtract like fractions, we add or subtract their numerators and retain the common denominator.

Example 1 : Simplify :

(i) $\frac{8}{15} + \frac{7}{15}$

(ii) $\frac{9}{10} + \frac{6}{10}$

(iii) $\frac{7}{8} - \frac{3}{8}$

(iv) $\frac{5}{9} - \frac{4}{9}$

Solution : (i) $\frac{8}{15} + \frac{7}{15} = \frac{8+7}{15} = \frac{15}{15} = 1$

(ii) $\frac{9}{10} + \frac{6}{10} = \frac{9+6}{10} = \frac{15}{10} = \frac{3}{2}$

(iii) $\frac{7}{8} - \frac{3}{8} = \frac{7-3}{8} = \frac{4}{8} = \frac{1}{2}$

(iv) $\frac{5}{9} - \frac{4}{9} = \frac{5-4}{9} = \frac{1}{9}$

Rule 2 : Fractions with different denominators can be added or subtracted by changing the given fractions to equivalent fractions having same denominator.

Example 2 : Simplify :

$$\begin{array}{llll} \text{(i)} \quad \frac{2}{9} + \frac{2}{3} & \text{(ii)} \quad \frac{3}{10} + \frac{1}{2} & \text{(iii)} \quad \frac{1}{6} + \frac{7}{12} & \text{(iv)} \quad \frac{2}{3} + \frac{1}{8} \\ \text{(v)} \quad \frac{5}{8} - \frac{1}{4} & \text{(vi)} \quad \frac{2}{3} - \frac{1}{4} & \text{(vii)} \quad \frac{3}{5} - \frac{1}{2} & \text{(viii)} \quad \frac{3}{4} - \frac{1}{10} \end{array}$$

Solution :

$$\begin{array}{ll} \text{(i)} \quad \frac{2}{9} + \frac{2}{3} = \frac{2}{9} + \left(\frac{2}{3} \times \frac{3}{3} \right) = \frac{2}{9} + \frac{6}{9} = \frac{2+6}{9} = \frac{8}{9} \\ \text{(ii)} \quad \frac{3}{10} + \frac{1}{2} = \frac{3}{10} + \left(\frac{1}{2} \times \frac{5}{5} \right) = \frac{3}{10} + \frac{5}{10} = \frac{3+5}{10} = \frac{8}{10} = \frac{4}{5} \\ \text{(iii)} \quad \frac{1}{6} + \frac{7}{12} = \left(\frac{1}{6} \times \frac{2}{2} \right) + \frac{7}{12} = \frac{2}{12} + \frac{7}{12} = \frac{2+7}{12} = \frac{9}{12} = \frac{3}{4} \\ \text{(iv)} \quad \frac{2}{3} + \frac{1}{8} = \left(\frac{2}{3} \times \frac{8}{8} \right) + \left(\frac{1}{8} \times \frac{3}{3} \right) = \frac{16}{24} + \frac{3}{24} = \frac{16+3}{24} = \frac{19}{24} \\ \text{(v)} \quad \frac{5}{8} - \frac{1}{4} = \frac{5}{8} - \left(\frac{1}{4} \times \frac{2}{2} \right) = \frac{5}{8} - \frac{2}{8} = \frac{5-2}{8} = \frac{3}{8} \\ \text{(vi)} \quad \frac{2}{3} - \frac{1}{4} = \left(\frac{2}{3} \times \frac{4}{4} \right) - \left(\frac{1}{4} \times \frac{3}{3} \right) = \frac{8}{12} - \frac{3}{12} = \frac{8-3}{12} = \frac{5}{12} \\ \text{(vii)} \quad \frac{3}{5} - \frac{1}{2} = \left(\frac{3}{5} \times \frac{2}{2} \right) - \left(\frac{1}{2} \times \frac{5}{5} \right) = \frac{6}{10} - \frac{5}{10} = \frac{6-5}{10} = \frac{1}{10} \\ \text{(viii)} \quad \frac{3}{4} - \frac{1}{10} = \left(\frac{3}{4} \times \frac{5}{5} \right) - \left(\frac{1}{10} \times \frac{2}{2} \right) = \frac{15}{20} - \frac{2}{20} = \frac{15-2}{20} = \frac{13}{20} \end{array}$$

Alternative Method for Addition and Subtraction

The addition and subtraction of fractions with different denominators can also be performed as follows :

$$\text{(i)} \quad \frac{7}{20} + \frac{13}{35}$$

L.C.M. of denominators 20 and 35 is 140.

$$\frac{7}{20} + \frac{13}{35} = \frac{7 \times 7 + 13 \times 4}{140}$$

$$\left[\begin{array}{l} 140 \div 20 = 7 \text{ and } 7 \times 7 = 49 \\ 140 \div 35 = 4 \text{ and } 13 \times 4 = 52 \end{array} \right]$$

$$= \frac{49 + 52}{140} = \frac{101}{140}$$

$$(ii) \frac{8}{9} - \frac{7}{12}$$

L.C.M. of denominators 9 and 12 is 36.

$$\frac{8}{9} - \frac{7}{12} = \frac{8 \times 4 - 7 \times 3}{36}$$

$$= \frac{32 - 21}{36} = \frac{11}{36}$$

$$\left[\begin{array}{l} 36 \div 9 = 4, 8 \times 4 = 32 \\ 36 \div 12 = 3, 7 \times 3 = 21 \end{array} \right]$$



EXERCISE 2.2

1. Find the sum of the following :

$$(i) \frac{5}{9} + \frac{7}{9}$$

$$(ii) \frac{2}{5} + \frac{3}{8}$$

$$(iii) \frac{5}{2} + \frac{4}{5}$$

$$(iv) \frac{3}{5} + \frac{7}{4}$$

$$(v) \frac{5}{8} + \frac{3}{4}$$

$$(vi) \frac{3}{4} + 1$$

$$(vii) \frac{2}{3} + \frac{3}{2}$$

$$(viii) \frac{4}{7} + \frac{11}{14}$$

2. Find the difference :

$$(i) \frac{5}{7} - \frac{3}{7}$$

$$(ii) \frac{11}{16} - \frac{7}{12}$$

$$(iii) \frac{7}{9} - \frac{2}{3}$$

$$(iv) \frac{15}{19} - \frac{12}{57}$$

$$(v) \frac{3}{7} - \frac{2}{5}$$

$$(vi) \frac{2}{3} - \frac{1}{8}$$

$$(vii) \frac{3}{4} - \frac{1}{2}$$

$$(viii) \frac{7}{10} - \frac{1}{2}$$

3. Find the sum :

$$(i) 3\frac{1}{9} + 1\frac{5}{6}$$

$$(ii) 2\frac{2}{3} + 3\frac{3}{5}$$

$$(iii) 12\frac{3}{5} + 1\frac{3}{5}$$

$$(iv) 10\frac{1}{9} + 6\frac{5}{12}$$

$$(v) 6\frac{1}{6} + 3\frac{2}{9}$$

$$(vi) 3\frac{1}{3} + 5\frac{2}{7}$$

4. Simplify :

$$(i) 5 - 3\frac{1}{5} - 1\frac{3}{5}$$

$$(ii) 4\frac{6}{7} - 2\frac{2}{3} - \frac{20}{21}$$

$$(iii) 5\frac{1}{6} - 3\frac{1}{4} + 2\frac{1}{3}$$

$$(iv) 10\frac{3}{5} + 2\frac{5}{6} - 5\frac{3}{5}$$

$$(v) \frac{2}{3} + \frac{11}{6} - 2\frac{1}{4}$$

$$(vi) 3 - \frac{7}{6} + \frac{11}{12}$$

5. What should be added to $\frac{11}{48}$ to get $\frac{13}{16}$?

6. Of $\frac{2}{3}$ and $\frac{7}{9}$, which is greater and by how much?

7. What should be added to $3\frac{1}{2}$ to get $8\frac{1}{4}$?

8. A rectangular field is $15\frac{2}{3}$ m long and $12\frac{4}{5}$ m wide. Find its perimeter.

9. The sum of two numbers is 7. If one of the numbers is $\frac{11}{7}$, find the other.
10. What number should be added to $\frac{5}{11}$, so as to get $\frac{7}{11}$?
11. A piece of wire is of length $8\frac{1}{4}$ m. It is cut into two pieces. The length of one piece is $4\frac{3}{4}$ m. What is the length of the other piece?
12. The sides of a triangle are $\frac{7}{2}$ cm, $\frac{11}{4}$ cm and $\frac{16}{5}$ cm. Find its perimeter.

MULTIPLICATION OF FRACTIONS

To multiply two fractions, we multiply their numerators to get the numerator of the product and multiply their denominators to get the denominator of the product.

Thus, if $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

i.e., **Product of Fractions = $\frac{\text{Product of their Numerators}}{\text{Product of their Denominators}}$**

Example 1 : Multiply and reduce the following fractions into their lowest form :

$$(i) \ 5 \times \frac{4}{9} \quad (ii) \ \frac{2}{3} \times 4 \quad (iii) \ 16 \times \frac{3}{4} \quad (iv) \ 25 \times \frac{4}{5}$$

Solution : (i) $5 \times \frac{4}{9} = \frac{5 \times 4}{1 \times 9} = \frac{5 \times 4}{1 \times 9} = \frac{20}{9} = 2\frac{2}{9}$

(ii) $\frac{2}{3} \times 4 = \frac{2 \times 4}{3 \times 1} = \frac{2 \times 4}{3 \times 1} = \frac{8}{3} = 2\frac{2}{3}$

(iii) $16 \times \frac{3}{4} = \frac{16 \times 3}{1 \times 4} = \frac{16 \times 3}{1 \times 4} = \frac{48}{4} = 12$

(iv) $25 \times \frac{4}{5} = \frac{25 \times 4}{1 \times 5} = \frac{25 \times 4}{1 \times 5} = \frac{100}{5} = 20$

Example 2 : Multiply and express each of the following as a mixed fraction :

$$(i) \ 5 \times 3\frac{2}{5} \quad (ii) \ 3\frac{1}{6} \times 15 \quad (iii) \ 7 \times 3\frac{1}{4} \quad (iv) \ 7 \times 6\frac{1}{3}$$

Solution : (i) $5 \times 3\frac{2}{5} = 5 \times \frac{17}{5} = 17$

(ii) $3\frac{1}{6} \times 15 = \frac{19}{6} \times 15 = \frac{95}{2} = 47\frac{1}{2}$

$$(iii) \quad 7 \times 3\frac{1}{4} = 7 \times \frac{13}{4} = \frac{91}{4} = 22\frac{3}{4}$$

$$(iv) \quad 7 \times 6\frac{1}{3} = 7 \times \frac{19}{3} = \frac{133}{3} = 44\frac{1}{3}$$

Example 3 : Simplify :

$$(i) \quad \frac{27}{4} \times \frac{8}{15} \times \frac{10}{3}$$

$$(ii) \quad 3\frac{3}{4} \times 3\frac{4}{7} \times 1\frac{2}{5}$$

Solution : (i) $\frac{27}{4} \times \frac{8}{15} \times \frac{10}{3} = \frac{\cancel{27}^3 \times \cancel{8}^2 \times \cancel{10}^2}{\cancel{4}_1 \times \cancel{15}_3 \times \cancel{3}_1} = 3 \times 2 \times 2 = 12$

$$(ii) \quad 3\frac{3}{4} \times 3\frac{4}{7} \times 1\frac{2}{5} = \frac{15}{4} \times \frac{25}{7} \times \frac{7}{5} = \frac{15 \times \cancel{25}^5 \times \cancel{7}^1}{4 \times \cancel{7}_1 \times \cancel{5}_1} = \frac{15 \times 5}{4} = \frac{75}{4} = 18\frac{3}{4}$$

Fraction as an Operator 'of'

Observe the adjacent figures. [Fig. (i)]

These are two exactly similar squares.

Each shaded portion represents $\frac{1}{2}$ of a square i.e., $\frac{1}{2}$ of 1.

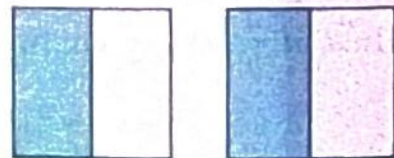


Fig. (i)

Both the shaded portions together will represent $\frac{1}{2}$ of 2.

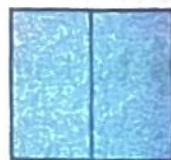


Fig. (ii)

If we combine the two shaded $\frac{1}{2}$ parts, it represents 1. [Fig. (ii)]

i.e., $\frac{1}{2}$ of 2 is 1

$\Rightarrow \frac{1}{2}$ of 2 = 1

Also, $\frac{1}{2} \times 2 = 1$

We see, that 'of' represents multiplication.

Here are some examples using an operator 'of'

$$(i) \quad \frac{1}{2} \text{ of } 4 = \frac{1}{2} \times 4 = 2$$

$$(ii) \quad \frac{1}{2} \text{ of } 5 = \frac{1}{2} \times 5 = \frac{5}{2}$$

$$(iii) \quad \frac{1}{2} \text{ of } 6 = \frac{1}{2} \times 6 = 3$$

$$(iv) \quad \frac{1}{2} \text{ of } 7 = \frac{1}{2} \times 7 = \frac{7}{2}$$

Example 4 : Find :

$$(i) \quad \frac{2}{3} \text{ of } 81$$

$$(ii) \quad \frac{5}{7} \text{ of } 98$$

$$(iii) \quad \frac{12}{13} \text{ of } 65$$

If the operator 'of' is used in any case like for an instance: $\frac{3}{4}$ of 2 then here 'of' means multiplication. It is telling us to do $\frac{3}{4} \times 2$ to have the answer of $\frac{3}{4} \times 2 = \frac{3}{2}$

Solution : (i) $\frac{2}{3}$ of 81 = $\frac{2}{3}$ of $\frac{81}{1}$

$$= \frac{2}{3} \times \frac{81}{1}$$

$$= \frac{2}{\cancel{3}} \times \frac{\cancel{81}^9}{1} = 2 \times 27 = 54$$

(ii) $\frac{5}{7}$ of 98 = $\frac{5}{7}$ of $\frac{98}{1}$

$$= \frac{5}{7} \times \frac{98}{1} = \frac{5 \times \cancel{98}^{14}}{\cancel{7} \times 1} = 5 \times 14 = 70$$

(iii) $\frac{12}{13}$ of 65 = $\frac{12}{13}$ of $\frac{65}{1}$

$$= \frac{12}{13} \times \frac{65}{1} = \frac{12 \times \cancel{65}^5}{\cancel{13} \times 1} = 12 \times 5 = 60$$



Puzzle

My father owns a small rose garden in Bangalore. My sister Rita has a share

of $\frac{3}{5}$ of it and my sister Gita has twice as much as myself. What fraction of the garden belongs to me?

Example 5 : Find :

(i) $\frac{1}{4}$ of a rupee

(ii) $\frac{3}{4}$ of a day

(iii) $\frac{7}{25}$ of a kg

(iv) $\frac{2}{3}$ of an hour

Solution : (i) $\frac{1}{4}$ of a rupee

We know, 1 rupee = 100 paise

$$\frac{1}{4} \text{ of a rupee} = \frac{1}{4} \text{ of 100 paise}$$

$$= \left(\frac{1}{4} \times 100 \right) \text{ paise} = \left(\frac{1 \times 100}{4} \right) \text{ paise} = 25 \text{ paise}$$

(ii) $\frac{3}{4}$ of a day

We know, 1 day = 24 hours

$$\frac{3}{4} \text{ of a day} = \frac{3}{4} \text{ of 24 hours}$$

$$= \left(\frac{3}{4} \times 24 \right) \text{ hours} = \left(\frac{3 \times \cancel{24}^6}{\cancel{4}^1} \right) \text{ hours} = 18 \text{ hours}$$

(iii) $\frac{7}{25}$ of a kg

We know, 1 kg = 1000 grams

$$\frac{7}{25} \text{ of a kg} = \frac{7}{25} \text{ of 1000 grams}$$

$$= \left(\frac{7}{25} \times 1000 \right) \text{ grams}$$

$$= \left(\frac{7 \times \cancel{1000}^{40}}{\cancel{1}25} \right) \text{ grams} = 280 \text{ grams}$$

(iv) $\frac{2}{3}$ of an hour

We know, 1 hour = 60 minutes

$$\frac{2}{3} \text{ of an hour} = \frac{2}{3} \text{ of 60 minutes}$$

$$= \left(\frac{2}{3} \times 60 \right) \text{ minutes} = \left(\frac{2 \times \cancel{60}^{20}}{\cancel{1}3} \right) \text{ minutes} = 40 \text{ minutes}$$

Example 6 : Which is greater : $\frac{3}{4}$ of 36 or $\frac{4}{5}$ of 35?

Solution :

$$\frac{3}{4} \text{ of } 36 = \frac{3}{4} \times 36 = \frac{3 \times \cancel{36}^9}{\cancel{1}4} = 27$$

$$\frac{4}{5} \text{ of } 35 = \frac{4}{5} \times 35 = \frac{4 \times \cancel{35}^7}{\cancel{1}5} = 28$$

Clearly, $28 > 27$

Hence, $\frac{4}{5}$ of 35 $>$ $\frac{3}{4}$ of 36

Value of the Products

We know that the product of two whole numbers is greater than each of the two whole numbers.

- Examples :** (i) $2 \times 3 = 6$, $6 > 2$ and $6 > 3$.
(ii) $3 \times 4 = 12$, $12 > 3$ and $12 > 4$.

Case I : When two proper fractions are multiplied, the value of their product is less than each of the two fractions.

- Examples :** (i) $\frac{1}{4} \times \frac{3}{5} = \frac{3}{20}$, $\frac{3}{20} < \frac{1}{4}$ and $\frac{3}{20} < \frac{3}{5}$ (ii) $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$, $\frac{8}{15} < \frac{2}{3}$ and $\frac{8}{15} < \frac{4}{5}$
(iii) $\frac{4}{7} \times \frac{1}{3} = \frac{4}{21}$, $\frac{4}{21} < \frac{4}{7}$ and $\frac{4}{21} < \frac{1}{3}$

Case II : When two improper fractions are multiplied, the value of their product is greater than each of the two fractions.

- Examples :** (i) $\frac{3}{2} \times \frac{5}{4} = \frac{15}{8}$; $\frac{15}{8} > \frac{3}{2}$ and $\frac{15}{8} > \frac{5}{4}$ (ii) $\frac{5}{3} \times \frac{7}{2} = \frac{35}{6}$; $\frac{35}{6} > \frac{5}{3}$ and $\frac{35}{6} > \frac{7}{2}$
(iii) $\frac{9}{4} \times \frac{7}{2} = \frac{63}{8}$; $\frac{63}{8} > \frac{9}{4}$ and $\frac{63}{8} > \frac{7}{2}$

Case III : When a proper fraction is multiplied by an improper fraction, the product obtained is less than the improper fraction but greater than the proper fraction.

Examples : (i) $\frac{2}{3} \times \frac{7}{5} = \frac{14}{15}$; $\frac{14}{15} < \frac{7}{5}$ and $\frac{14}{15} > \frac{2}{3}$ (ii) $\frac{3}{4} \times \frac{9}{2} = \frac{27}{8}$; $\frac{27}{8} < \frac{9}{2}$ and $\frac{27}{8} > \frac{3}{4}$

(iii) $\frac{1}{4} \times \frac{5}{2} = \frac{5}{8}$; $\frac{5}{8} < \frac{5}{2}$ and $\frac{5}{8} > \frac{1}{4}$

WORD PROBLEMS

Example 1 : Vishakha reads a book for $1\frac{4}{5}$ hours everyday. She reads the entire book in 10 days. How many hours in all does she take to read the book?

Solution : In one day, she reads for $= 1\frac{4}{5}$ hours $= \frac{9}{5}$ hours

In 10 days, she reads for $= \frac{9}{5} \times 10 = 18$ hours

$\therefore$ 18 hours were required by her to read the book.

Example 2 : Shikha's age is $\frac{2}{5}$ of the age of her brother. If her brother's age is 35 years, then how old is Shikha?

Solution : Shikha's age $= \frac{2}{5}$ of the age of her brother (given)

$= \frac{2}{5}$ of 35 years (Her brother's age = 35 years)

$= \left(\frac{2}{5} \times 35 \right)$ years = 14 years

$\therefore$ Shikha is 14 years old.

Example 3 : Find the perimeter and area of a square field whose side is $5\frac{1}{4}$ m long.

Solution : Each side of the square $= 5\frac{1}{4}$ m $= \frac{21}{4}$ m

Perimeter of the square $= 4 \times (\text{side})$

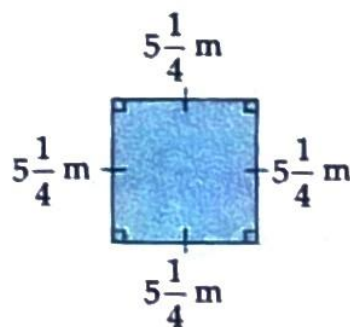
$= 4 \times \frac{21}{4}$ m = 21 m

Area of the square $= (\text{side})^2$

$= \left(\frac{21}{4} \right)^2$ sq. m $= \left(\frac{21}{4} \times \frac{21}{4} \right)$ sq. m

$= \frac{441}{16}$ sq. m $= 27\frac{9}{16}$ sq. m

Hence, the perimeter of the square is 21 m and area is $27\frac{9}{16}$ sq. m.



Example 4 : Vinay covers $7\frac{1}{2}$ km in one hour. How much distance will he cover in $2\frac{4}{5}$ hours?

Solution : $7\frac{1}{2}$ km = $\frac{15}{2}$ km; $2\frac{4}{5}$ hours = $\frac{14}{5}$ hours;

Distance covered in one hour = $\frac{15}{2}$ km

$\therefore$ Distance covered in $\frac{14}{5}$ hours = $\frac{15^3}{2^7} \times \frac{14^7}{5^8} = 21$ km

Hence, Vinay will cover 21 km in $2\frac{4}{5}$ hours.



EXERCISE 2.3

1. Multiply and reduce to lowest form and convert into a mixed fraction :

- | | | | |
|-----------------------------|------------------------------|-------------------------------|--------------------------------|
| (i) $9 \times \frac{2}{5}$ | (ii) $5 \times \frac{3}{7}$ | (iii) $13 \times \frac{1}{4}$ | (iv) $16 \times \frac{3}{4}$ |
| (v) $15 \times \frac{2}{5}$ | (vi) $11 \times \frac{3}{5}$ | (vii) $5 \times \frac{1}{3}$ | (viii) $\frac{5}{3} \times 27$ |

2. Multiply and express as a mixed fraction :

- | | | |
|------------------------------|------------------------------|-------------------------------|
| (i) $6 \times 2\frac{1}{9}$ | (ii) $5\frac{1}{4} \times 6$ | (iii) $2\frac{3}{5} \times 6$ |
| (iv) $9 \times 3\frac{1}{4}$ | (v) $7 \times 2\frac{1}{3}$ | (vi) $10 \times 6\frac{1}{4}$ |

3. Find :

- | | | |
|-------------------------------------|--------------------------------------|---------------------------------------|
| (i) $\frac{1}{2}$ of $3\frac{2}{9}$ | (ii) $\frac{1}{3}$ of $\frac{15}{7}$ | (iii) $\frac{1}{3}$ of $3\frac{2}{9}$ |
|-------------------------------------|--------------------------------------|---------------------------------------|

4. Multiply and reduce to lowest form (if possible) :

- | | | | |
|---------------------------------------|--|---|--|
| (i) $\frac{3}{5} \times 2\frac{2}{5}$ | (ii) $\frac{3}{5} \times \frac{5}{6}$ | (iii) $\frac{1}{4} \times \frac{32}{3}$ | (iv) $\frac{11}{3} \times \frac{4}{5}$ |
| (v) $2\frac{3}{5} \times \frac{3}{7}$ | (vi) $7\frac{3}{4} \times \frac{8}{5}$ | (vii) $\frac{3}{2} \times 7\frac{2}{3}$ | (viii) $\frac{5}{6} \times 2\frac{2}{7}$ |

5. Which is greater ?

- | | |
|--|---|
| (i) $\frac{2}{3}$ of $\frac{1}{5}$ or $\frac{3}{5}$ of $\frac{5}{6}$ | (ii) $\frac{4}{5}$ of $\frac{5}{6}$ or $\frac{2}{3}$ of $\frac{4}{5}$ |
|--|---|

6. Find :

- | | | |
|-------------------------------|--------------------------------|------------------------------|
| (i) $\frac{3}{4}$ of a year | (ii) $\frac{2}{5}$ of a rupee | (iii) $\frac{1}{3}$ of a day |
| (iv) $\frac{1}{4}$ of an hour | (v) $\frac{11}{25}$ of a litre | (vi) $\frac{3}{8}$ of a kg |

7. Priyanka has 24 apples. $\frac{3}{4}$ of these are found to be rotten. Find the number of rotten apples.
8. A rectangular park is $21\frac{3}{7}$ m long and $12\frac{3}{5}$ m wide. What is the area of the park?
9. Apples are sold at ₹ $15\frac{3}{4}$ per kg. Find the cost of $6\frac{1}{2}$ kg of apples.
10. Rohit covers $2\frac{2}{5}$ km in one hour. How much distance will he cover in $3\frac{1}{3}$ hours?
11. There are 48 students in a class and $\frac{2}{3}$ of them are boys. How many girls are there in the class?
12. A car runs 18 km in 1 litre of petrol. How much distance will it cover in $3\frac{1}{3}$ litres of petrol?
13. Varad reads $\frac{1}{4}$ part of a book in one hour. How much part of the book will he read in $3\frac{1}{2}$ hours?
14. A book contains 320 pages. Vishal reads $\frac{3}{5}$ of the book. How many pages are left to be read?
15. A tin contains 21 kg of sugar. If $\frac{2}{3}$ of it has been consumed, how much sugar is left in the tin?

MULTIPLICATIVE INVERSE (RECIPROCAL) OF A FRACTION

If we multiply 3 by $\frac{1}{3}$ i.e., $3 \times \frac{1}{3}$ we get 1.

Similarly, $\frac{2}{3}$ has to be multiplied by $\frac{3}{2}$ to get 1.

$\frac{1}{3}$ is the reciprocal of 3 and 3 is the reciprocal of $\frac{1}{3}$.

$\frac{2}{3}$ is the reciprocal of $\frac{3}{2}$ and $\frac{3}{2}$ is the reciprocal of $\frac{2}{3}$.

Thus, if the product of two fractions is 1, then one fraction is the reciprocal or multiplicative inverse of the other.

Clearly, the reciprocal of a non-zero fraction $\frac{a}{b}$ is the fraction $\frac{b}{a}$.

Example 1 : Find the reciprocal of :

(i) $\frac{2}{9}$

(ii) $\frac{3}{8}$

(iii) $\frac{4}{3}$

Solution : (i) The reciprocal i.e., multiplicative inverse of $\frac{2}{9}$ is $\frac{1}{\frac{2}{9}} = \frac{9}{2}$.

(ii) The reciprocal of $\frac{3}{8}$ is $\frac{1}{\frac{3}{8}} = \frac{8}{3}$.

(iii) The reciprocal of $\frac{4}{3}$ is $\frac{1}{\frac{4}{3}} = \frac{3}{4}$.

DIVISION OF FRACTIONS

We know that division is the inverse of multiplication.

If we divide 6 by 3, then $6 \div 3 = 6 \times \frac{1}{3} = 2$

Similarly, $\frac{4}{5} \div \frac{2}{3} = \frac{4}{5} \times \left(\frac{1}{\frac{2}{3}}\right) = \frac{4}{5} \times \frac{3}{2} = \frac{6}{5}$

fractions

Thus, we observe that if we divide a fraction by another fraction, the quotient is the product of first fraction and reciprocal of second fraction.

$$\therefore \frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Example 1 : Divide :

(i) $\frac{7}{9}$ by $\frac{2}{3}$

(ii) 35 by $\frac{7}{5}$

(iii) 49 by $2\frac{3}{7}$

(iv) $\frac{13}{5}$ by 11

Solution : (i) $\frac{7}{9} \div \frac{2}{3} = \frac{7}{9} \times \frac{3}{2}$

(Reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$)

$$= \frac{7}{9} \times \frac{3^1}{2} = \frac{7}{6} = 1\frac{1}{6}$$

(ii) $35 \div \frac{7}{5} = \frac{35}{1} \div \frac{7}{5} = \frac{35}{1} \times \frac{5}{7}$

(Reciprocal of $\frac{7}{5}$ is $\frac{5}{7}$)

$$= \frac{35^1}{1} \times \frac{5}{7} = 25$$

(iii) $49 \div 2\frac{3}{7} = \frac{49}{1} \div \frac{17}{7}$

$$= \frac{49}{1} \times \frac{7}{17}$$

(Reciprocal of $\frac{17}{7}$ is $\frac{7}{17}$)

$$= \frac{343}{17} = 20\frac{3}{17}$$

(iv) $\frac{13}{5} \div 11 = \frac{13}{5} \times \frac{1}{11}$

(Reciprocal of 11 is $\frac{1}{11}$)

$$= \frac{13}{55}$$

Example 2 : Simplify :

$$(i) 48 + \frac{8}{9}$$

$$(ii) 45 + 2\frac{3}{5}$$

$$(iii) \frac{4}{5} + \frac{2}{3}$$

$$(iv) \frac{25}{32} + \frac{5}{8}$$

$$(v) \frac{5}{9} + 2\frac{7}{9}$$

Solution :

$$(i) 48 + \frac{8}{9} = \frac{48}{1} + \frac{8}{9} = \frac{48^6}{1} \times \frac{9}{1^8} = 54 \quad \left(\text{Reciprocal of } \frac{8}{9} \text{ is } \frac{9}{8} \right)$$

$$(ii) 45 + 2\frac{3}{5} = \frac{45}{1} + \frac{13}{5} = \frac{45}{1} \times \frac{5}{13} = \frac{225}{13} = 17\frac{4}{13} \quad \left(\text{Reciprocal of } \frac{13}{5} \text{ is } \frac{5}{13} \right)$$

$$(iii) \frac{4}{5} + \frac{2}{3} = \frac{4}{5} \times \frac{3}{2} = \frac{4^2}{5} \times \frac{3}{2^1} = \frac{6}{5} = 1\frac{1}{5} \quad \left(\text{Reciprocal of } \frac{2}{3} \text{ is } \frac{3}{2} \right)$$

$$(iv) \frac{25}{32} + \frac{5}{8} = \frac{25}{32} \times \frac{8}{5} = \frac{25^5}{32^4} \times \frac{8^1}{5^1} = \frac{5}{4} = 1\frac{1}{4} \quad \left(\text{Reciprocal of } \frac{5}{8} \text{ is } \frac{8}{5} \right)$$

$$(v) \frac{5}{9} + 2\frac{7}{9} = \frac{5}{9} + \frac{25}{9} = \frac{5^1}{9^1} \times \frac{9^1}{25^5} = \frac{1}{5} \quad \left(\text{Reciprocal of } \frac{25}{9} \text{ is } \frac{9}{25} \right)$$

Example 3 : Divide :

$$(i) 2\frac{2}{9} \div \frac{25}{27}$$

$$(ii) 3\frac{1}{3} \div 2\frac{1}{6}$$

$$(iii) 2\frac{2}{9} \div 1\frac{2}{3}$$

$$(iv) 2\frac{2}{3} \div 1\frac{5}{9}$$

Solution :

$$(i) 2\frac{2}{9} \div \frac{25}{27} = \frac{20}{9} \div \frac{25}{27} = \frac{20^4}{9} \times \frac{27^3}{25^5} = \frac{12}{5} = 2\frac{2}{5}$$

$$(ii) 3\frac{1}{3} \div 2\frac{1}{6} = \frac{10}{3} \div \frac{13}{6} = \frac{10}{3} \times \frac{6^2}{13} = \frac{20}{13} = 1\frac{7}{13}$$

$$(iii) 2\frac{2}{9} \div 1\frac{2}{3} = \frac{20}{9} \div \frac{5}{3} = \frac{20^4}{9} \times \frac{3^1}{5^1} = \frac{4}{3} = 1\frac{1}{3}$$

$$(iv) 2\frac{2}{3} \div 1\frac{5}{9} = \frac{8}{3} \div \frac{14}{9} = \frac{8^4}{3} \times \frac{9^3}{14^7} = \frac{12}{7} = 1\frac{5}{7}$$

Example 4 : The product of two numbers is $1\frac{1}{5}$. If one of the numbers is $\frac{4}{5}$, find the other number.

Solution : Let the other number be x .

$$\therefore x \times \frac{4}{5} = 1\frac{1}{5}$$

$$\Rightarrow x \times \frac{4}{5} = \frac{6}{5}$$

$$\Rightarrow x = \frac{6}{5} \div \frac{4}{5}$$

$$\Rightarrow x = \frac{6^3}{5} \times \frac{5^1}{4^1} = \frac{3}{2} = 1\frac{1}{2}$$

The other number is $1\frac{1}{2}$.

Example 5 : What number should be multiplied with $\frac{25}{32}$ to get the product $\frac{5}{8}$?

Solution : Let the required number be x .

$$\therefore x \times \frac{25}{32} = \frac{5}{8}$$

$$\Rightarrow x = \frac{5}{8} \div \frac{25}{32}$$

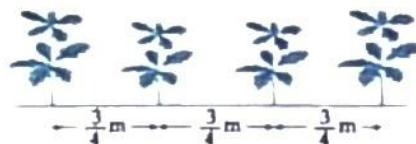
$$\Rightarrow x = \frac{5}{8} \times \frac{32}{25}$$

$$\Rightarrow x = \frac{4}{5}$$

The required number is $\frac{4}{5}$.

VALUE CORNER

Shubda plants 4 saplings, in a row, in her garden. The distance between two adjacent saplings is $\frac{3}{4}$ m. Find the distance between the first and the last sapling.



Why is planting saplings important for us?

Example 6 : If the cost of 17 m of cloth is ₹ $77\frac{5}{7}$, find its cost per metre.

Solution : Cost of 17 m of cloth = ₹ $77\frac{5}{7} = ₹ \frac{544}{7}$

$$\begin{aligned} \Rightarrow \text{Cost of 1 m of cloth} &= ₹ \left(\frac{544}{7} \div 17 \right) \\ &= ₹ \left(\frac{544}{7} \times \frac{1}{17} \right) \\ &= ₹ \frac{32}{7} = ₹ 4\frac{4}{7} \end{aligned}$$

Hence, the cost of cloth is ₹ $4\frac{4}{7}$ per metre.



EXERCISE 2.4

1. Divide :

(i) $\frac{14}{9}$ by 7

(ii) $\frac{15}{27}$ by 5

(iii) $\frac{49}{6}$ by 7

(iv) $\frac{48}{39}$ by 8

(v) $\frac{18}{25}$ by $\frac{3}{50}$

(vi) $\frac{5}{7}$ by 10

(vii) 24 by $\frac{8}{3}$

2. Evaluate :

(i) $3\frac{3}{5} + 2\frac{2}{5}$

(ii) $6\frac{7}{8} + \frac{11}{16}$

(iii) $40 + 6\frac{2}{9}$

(iv) $6\frac{2}{5} + 8$

(v) $2\frac{4}{7} + 1\frac{5}{7}$

(vi) $\frac{95}{6} + 20$

3. Evaluate :

(i) $5\frac{1}{2} + 2\frac{1}{5}$

(ii) $4\frac{1}{2} + \frac{3}{10}$

(iii) $\frac{1}{6} + \frac{2}{3}$

(iv) $\frac{27}{5} + \frac{3}{2}$

(v) $1\frac{7}{104} + \frac{11}{13}$

(vi) $18\frac{3}{4} + 6\frac{1}{4}$

(vii) $\frac{4}{9} + \frac{5}{6}$

4. Write down the reciprocal of each of the following fractions :

(i) $\frac{1}{13}$

(ii) $12\frac{2}{5}$

(iii) 9

(iv) $\frac{6}{7}$

5. The cost of $4\frac{1}{4}$ kg of sugar is ₹ 68. Find the cost of 1 kg.

6. The product of a fraction and the sum of $2\frac{3}{4}$ and $3\frac{1}{2}$ is 5. Find the fraction.

7. Find a number such that its two-fifths is 12.

8. The product of two numbers is $\frac{16}{9}$. If one of the numbers is $\frac{5}{2}$, find the other number.

9. A rope of length $3\frac{3}{5}$ m has been divided into six pieces of equal length. What is the length of each piece?

10. If the cost of a notebook is ₹ $6\frac{1}{3}$, how many notebooks can be purchased for ₹ $126\frac{2}{3}$?

11. By what number should $10\frac{2}{3}$ be divided to obtain 20?

12. By what number should $5\frac{4}{7}$ be divided to get 6?

13. Find a number such that its two-thirds is equal to $\frac{5}{4}$.



FACTS TO REMEMBER

- The numbers of the form $\frac{a}{b}$, where a and b are whole numbers and $b \neq 0$ are called fractions. Here, a is called the numerator and b is called the denominator of fraction.
- A fraction whose numerator is less than the denominator is called a proper fraction.
- A fraction whose numerator is more than or equal to the denominator is called an improper fraction.
- A combination of whole number and a proper fraction is called a mixed fraction.
- The reciprocal of a fraction is obtained by interchanging the numerator and denominator of the fraction.
- Sum of like fractions = $\frac{\text{Sum of their numerators}}{\text{Common denominator}}$

- Difference of like fractions = $\frac{\text{Difference of their numerators}}{\text{Common denominator}}$
- For adding or subtracting unlike fractions, change them into equivalent like fractions and then add or subtract.
- If $\frac{a}{b}$ and $\frac{c}{d}$ are two fractions, then $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$.
- To divide a fraction by another fraction, multiply the first fraction by the reciprocal of the second fraction. Thus, $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$.



MULTIPLE CHOICE QUESTIONS

Choose the correct alternatives in each of the following :

1. Which of the following is a reducible fraction?

(a) $\frac{79}{26}$

(b) $\frac{41}{17}$

(c) $\frac{105}{112}$

(d) $\frac{91}{15}$

2. Which of the following is an improper fraction?

(a) $\frac{3}{5}$

(b) $\frac{8}{9}$

(c) $\frac{11}{13}$

(d) $\frac{9}{8}$

3. Reciprocal of $2\frac{3}{5}$ is

(a) $5\frac{3}{2}$

(b) $3\frac{2}{5}$

(c) $2\frac{3}{5}$

(d) $\frac{5}{13}$

4. $25 \div \frac{1}{5} = ?$

(a) 5

(b) $\frac{1}{5}$

(c) 125

(d) $\frac{1}{125}$

5. Which of the following statements is true?

(a) $\frac{29}{6} = \frac{43}{12}$

(b) $\frac{29}{6} > \frac{43}{2}$

(c) $\frac{29}{6} < \frac{43}{2}$

(d) $\frac{29}{6} = \frac{52}{12}$

6. To get 40, $6\frac{2}{9}$ should be multiplied with

(a) $7\frac{3}{6}$

(b) $6\frac{3}{7}$

(c) $3\frac{6}{7}$

(d) $6\frac{2}{7}$



MENTAL MATHS CORNER

Fill in the blanks :

1. Reciprocal of $\frac{9}{2}$ is
2. $\frac{2}{3}$ of 15 litres is litres.
3. $7\frac{3}{8}$ as an improper fraction can be written as
4. $\frac{103}{6}$ as a mixed fraction can be written as
5. $\frac{9}{5}$ of $\frac{15}{27} =$
6. $1 + \frac{3}{5} =$
7. The product of two fractions is $\frac{1}{4}$. If one of the fractions is $\frac{2}{3}$, then the other fraction is
8. $\frac{5}{8}$ of a kilogram = grams.
9. $7\frac{5}{9}$ must be multiplied with to get $11\frac{1}{3}$.
10. $\frac{2}{3} \times \frac{4}{9} \times 0 \times \frac{3}{5} =$
11. The non-zero numbers whose product with each other is 1 are called
12. (i) $\frac{3}{4} + 3 =$ (ii) $15 + \frac{3}{5} =$ (iii) $\frac{2}{4} + \frac{1}{2} =$
(iv) $\frac{3}{2} - 1 =$ (v) $\frac{2}{5} + 1\frac{1}{2} =$ (vi) $\frac{5}{2} - 2 =$
13. The value of the product of two improper fractions is than each of the two fractions.
14. 4 times $\frac{1}{4}$ is equal to



REVIEW EXERCISE

1. Convert the following into mixed fractions :

(i) $\frac{37}{5}$

(ii) $\frac{77}{8}$

(iii) $\frac{105}{9}$

2. Convert the following into improper fractions :

(i) $3\frac{5}{7}$

(ii) $2\frac{6}{11}$

(iii) $8\frac{3}{5}$

(iv) $9\frac{4}{11}$

3. Arrange in ascending order : $\frac{5}{12}, \frac{3}{4}, \frac{7}{8}, \frac{13}{24}$

4. Write four equivalent fractions of $\frac{6}{7}$.

5. Find the sum of the following :

(i) $\frac{5}{12} + \frac{1}{3}$

(ii) $4\frac{1}{3} + 5\frac{2}{5}$

(iii) $8 + \frac{16}{7}$

(iv) $2\frac{1}{4} + 12$

6. Subtract the following :

(i) $\frac{1}{15}$ from $\frac{3}{10}$

(ii) $\frac{7}{9}$ from $1\frac{5}{18}$

7. Find :

(i) $\frac{2}{13}$ of 117

(ii) $\frac{3}{7}$ of a week

(iii) $3\frac{3}{4}$ of $8\frac{2}{5}$

8. Simplify : $\frac{12}{25} \times \frac{10}{18} \times \frac{15}{8}$

9. Multiply :

(i) $3\frac{5}{7} \times 4\frac{2}{3}$

(ii) $2\frac{1}{12} \times 1\frac{1}{15}$

10. Divide :

(i) $5\frac{3}{5}$ by $2\frac{1}{10}$

(ii) $7\frac{1}{2}$ by $4\frac{2}{7}$

11. Divide the sum of the fractions $\frac{3}{2}$ and $\frac{5}{7}$ by their difference.

12. In a class, there are 45 students. If $\frac{3}{5}$ of the students are boys, find the number of boys in the class.

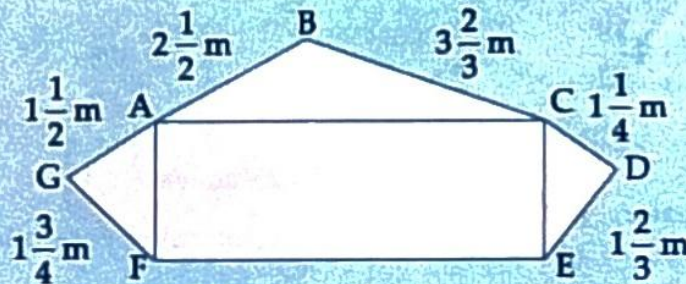
13. A cloth merchant sells cloth at the rate of ₹ $23\frac{3}{5}$ per metre. What is the cost of $6\frac{3}{7}$ m of cloth?

14. Rajni's monthly income is ₹ 21000. Her monthly expenses are $\frac{6}{7}$ of her income. How much does she save every month?



HOTS QUESTIONS

1. The perimeter of the given figure is 18 m and the area of rectangular part ACEF is $8\frac{1}{2} \text{ m}^2$. Find the length and breadth of the rectangular part.



2. Find the fraction which is as much greater than $\frac{4}{7}$ as it is less than $\frac{5}{6}$.



ANSWERS

Exercise 2.1

- (i) $\frac{3}{8}$ (ii) $\frac{5}{7}$ (iii) $\frac{1}{4}$ (iv) $\frac{3}{4}$
- (i) $5\frac{1}{3}$ (ii) $4\frac{4}{7}$ (iii) $18\frac{9}{13}$ (iv) $25\frac{16}{17}$
- (i) $\frac{4}{9}, \frac{5}{12}, \frac{7}{24}, \frac{5}{27}$ (ii) $\frac{3}{4}, \frac{5}{7}, \frac{3}{8}, \frac{5}{14}$
- (i) $\frac{7}{12}, \frac{3}{4}, \frac{5}{6}, \frac{11}{12}$ (ii) $\frac{8}{15}, \frac{13}{20}, \frac{7}{10}, \frac{4}{5}$
(iii) $\frac{16}{35}, \frac{6}{10}, \frac{9}{14}, \frac{4}{5}$ (iv) $\frac{2}{15}, \frac{1}{5}, \frac{3}{7}, \frac{7}{10}$
- (i) $\frac{11}{4}$ (ii) $\frac{33}{10}$ (iii) $\frac{20}{3}$ (iv) $\frac{17}{6}$ (v) $\frac{32}{5}$
- (i) Yes (ii) Yes (iii) No (iv) No (v) Yes (vi) Yes

Exercise 2.2

- (i) $1\frac{1}{3}$ (ii) $\frac{31}{40}$ (iii) $3\frac{3}{10}$ (iv) $2\frac{7}{20}$ (v) $1\frac{3}{8}$
(vi) $1\frac{3}{4}$ (vii) $2\frac{1}{6}$ (viii) $1\frac{5}{14}$

- (i) $\frac{2}{7}$ (ii) $\frac{5}{48}$ (iii) $\frac{1}{9}$ (iv) $\frac{11}{19}$ (v) $\frac{1}{35}$
(vi) $\frac{13}{24}$ (vii) $\frac{1}{4}$ (viii) $\frac{1}{5}$
- (i) $4\frac{17}{18}$ (ii) $6\frac{4}{15}$ (iii) $14\frac{1}{5}$ (iv) $16\frac{19}{36}$ (v) $9\frac{7}{18}$
(vi) $8\frac{13}{21}$
- (i) $\frac{1}{5}$ (ii) $1\frac{5}{21}$ (iii) $4\frac{1}{4}$ (iv) $7\frac{5}{6}$ (v) $\frac{1}{4}$ (vi) $2\frac{3}{4}$
- $\frac{7}{12}$ 6. $\frac{7}{9}, \frac{1}{9}$ 7. $4\frac{3}{4}$ 8. $56\frac{14}{15} \text{ m}$
- $5\frac{3}{7}$ 10. $\frac{2}{11}$ 11. $\frac{7}{2} \text{ m}$ 12. $9\frac{9}{20} \text{ cm}$

Exercise 2.3

- (i) $3\frac{3}{5}$ (ii) $2\frac{1}{7}$ (iii) $3\frac{1}{4}$ (iv) 12 (v) 6 (vi) $6\frac{3}{5}$
(vii) $1\frac{2}{3}$ (viii) 45

2. (i) $12\frac{2}{3}$ (ii) $31\frac{1}{2}$ (iii) $15\frac{3}{5}$ (iv) $29\frac{1}{4}$

(v) $16\frac{1}{3}$ (vi) $62\frac{1}{2}$

3. (i) $1\frac{11}{18}$ (ii) $\frac{5}{7}$ (iii) $1\frac{2}{27}$

4. (i) $1\frac{11}{25}$ (ii) $\frac{1}{2}$ (iii) $2\frac{2}{3}$ (iv) $2\frac{14}{15}$ (v) $1\frac{4}{35}$

(vi) $12\frac{2}{5}$ (vii) $11\frac{1}{2}$ (viii) $1\frac{19}{21}$

5. (i) $\frac{3}{5}$ of $\frac{5}{6}$ (ii) $\frac{4}{5}$ of $\frac{5}{6}$

6. (i) 9 months (ii) 40 paise (iii) 8 hours
(iv) 15 minutes (v) 440 ml (vi) 375 g

7. 18 8. 270 sq. m 9. ₹ $102\frac{3}{8}$

10. 8 km 11. 16 girls 12. 60 km

13. $\frac{7}{8}$ 14. 128 pages 15. 7 kg

Exercise 2.4

1. (i) $\frac{2}{9}$ (ii) $\frac{1}{9}$ (iii) $1\frac{1}{6}$ (iv) $\frac{2}{13}$ (v) 12

(vi) $\frac{1}{14}$ (vii) 9

2. (i) $1\frac{1}{2}$ (ii) 10 (iii) $6\frac{3}{7}$ (iv) $\frac{4}{5}$ (v) $1\frac{1}{2}$ (vi) $\frac{19}{24}$

3. (i) $2\frac{1}{2}$ (ii) 15 (iii) $\frac{1}{4}$ (iv) $\frac{18}{5}$ (v) $1\frac{23}{88}$ (vi) 3

(vii) $\frac{8}{15}$

4. (i) 13 (ii) $\frac{5}{62}$ (iii) $\frac{1}{9}$ (iv) $\frac{7}{6}$

5. ₹ 16 6. $\frac{4}{5}$ 7. 30 8. $\frac{32}{45}$

9. $\frac{3}{5}$ m 10. 20 11. $\frac{8}{15}$ 12. $\frac{13}{14}$

13. $1\frac{7}{8}$

Multiple Choice Questions

1. (c) 2. (d) 3. (d) 4. (c) 5. (c)
6. (b)

Mental Maths Corner

1. $\frac{2}{9}$ 2. 10 litres 3. $\frac{59}{8}$ 4. $17\frac{1}{6}$

5. 1 6. $\frac{5}{3}$ 7. $\frac{3}{8}$

8. 625 grams 9. $1\frac{1}{2}$ 10. 0

11. Reciprocal of each other

12. (i) $\frac{1}{4}$ (ii) 25 (iii) 1 (iv) $\frac{1}{2}$ (v) $\frac{4}{15}$ (vi) $\frac{1}{2}$

13. greater 14. 1

Review Exercise

1. (i) $7\frac{2}{5}$ (ii) $9\frac{5}{8}$ (iii) $11\frac{2}{3}$

2. (i) $\frac{26}{7}$ (ii) $\frac{28}{11}$ (iii) $\frac{43}{5}$ (iv) $\frac{103}{11}$

3. $\frac{5}{12}, \frac{13}{24}, \frac{3}{4}, \frac{7}{8}$ 4. $\frac{12}{14}, \frac{18}{21}, \frac{24}{28}, \frac{30}{35}$

5. (i) $\frac{3}{4}$ (ii) $9\frac{11}{15}$ (iii) $10\frac{2}{7}$ (iv) $14\frac{1}{4}$

6. (i) $\frac{7}{30}$ (ii) $\frac{1}{2}$ 7. (i) 18 (ii) 3 days (iii) $31\frac{1}{2}$

8. $\frac{1}{2}$ 9. (i) $17\frac{1}{3}$ (ii) $2\frac{2}{9}$ 10. (i) $2\frac{2}{3}$ (ii) $1\frac{3}{4}$

11. $2\frac{9}{11}$ 12. 27 13. ₹ $151\frac{5}{7}$ 14. ₹ 3000

HOTS Questions

1. $5\frac{2}{3}$ m, $1\frac{1}{2}$ m 2. $\frac{59}{84}$